HOPFIELD NETWORK / AUTOENCODER / ATTRACTOR

GOAL (1) MEMORY (CONTENT ADDRESSABLE MEM vs. INDEXED)

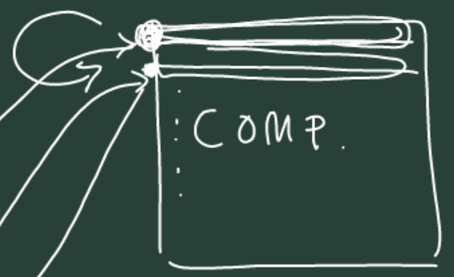
(2) HEBBIAN LEARNING.

CAT [0 1 1 1 0]

MEOW [0 1 1 0 0]

⋮

INDEXED MEMORY.



"agency"

- Depth.

Relationship state (phys.) vs. content (psy.) is arbitrary.

CONTENT ADDRESSABLE MEMORY

- If we change the neuron "used" or its state, we change the content.
  - Implicit learning.
  - False memory?
  - Constrained (more) exp.
  - Fuzzy categories.
    - analogical ⇒ higher resolution
    - quantum-like

INDEXED MEMORY
↳ STATE ≠ CONTENT

CAM
↳ STATE ~ CONTENT

DONALD HEBB

FIRE TOGETHER, WIRE TOGETHER.

e.g. NMDA receptor, synaptic, activity-dependant learn$^{\underline{g}}$.

- LONG-TERM POTENTIAT$^{\underline{n}}$
- LSTP
- LTDep.

① Content Addressable Memory

Let $S = 1, 2, \ldots, n$ sets of distinct states, item features that we wish to encode & retrieve based on item features $\boxed{\text{themselves}}$ ($N = \text{length}(S)$, i.e, no. of states).

SELF, "AUTO"

i.e. autoencode $\longrightarrow$ auto-based retrieval.

E.g. Feature space: {Leaves, Wood, Fur, Tail, Meows}

S

1 :	CAT = [	0	0	1	1	1]
2 :	DOG = [	0	0	1	1	0]
3 :	TREE = [	1	1	0	0	0]
4 :	COTTON PLANT = [	1	1	1	0	0]

nb: Delocalizat$^{\underline{n}}$ of Representat$^{\underline{n}}$

s.t. Given "leaves" (probe) i.e. [1 0 0 0 0] we can retrieve "TREE" i.e. [1 1 0 0 0] using a neural network.

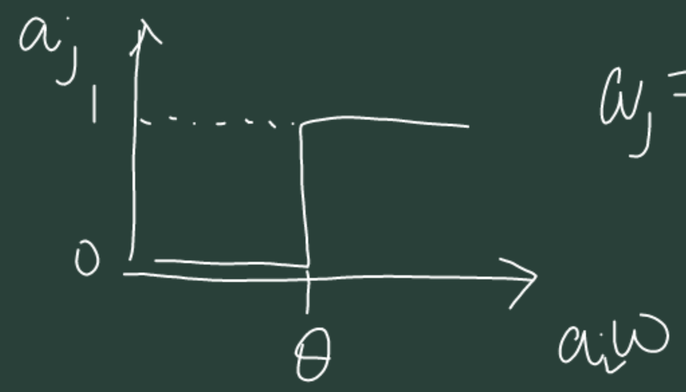
DATA TABLE roughly.

② ACTIVATION FUNCTION

$$a_j = \begin{cases} 1 & \text{if } \sum_i a_i w_{ik)r} \geq \theta \\ 0 & \sum_i a_i w_{ik)r} < \theta \end{cases} \qquad (1)$$

connecting weights.

- Threshold function.
- Binary ReLU (regularized linear unit).



$$a_j = \max(\text{sign}(a_i w - \theta), 0)$$

③ LEARNING RULE

UPDATING FUNCTION.

- Hebbian Learning Rule.

$$w_{ik)r} = \sum_s (2a_{ik}^s - 1)(2a_{jr}^s - 1),$$

$$w_{ik)r} = 0 \text{ for } k = r$$

$$(2)$$

VISUALIZE

CAT	DOG	TREE	COTTONP.
1	2	3	4

$$S \in \begin{bmatrix} a_{i_1} & a_{i_2} & & \\ a_{i_2} & \vdots & & \\ a_{i_3} & \vdots & & \\ a_{i_4} & \vdots & & \\ a_{i_5} & a_{i_5} & & \end{bmatrix}$$

$$\downarrow$$

$$\vec{a}_i$$

length(s).

$N_s = 4$

$k \in \{1,2,3,4,5\}$ Input nodes.

$r \in \{1,2,3,4,5\}$ output nodes.

Input & output here refers to prior and posterior (subsequent) states of node activity.

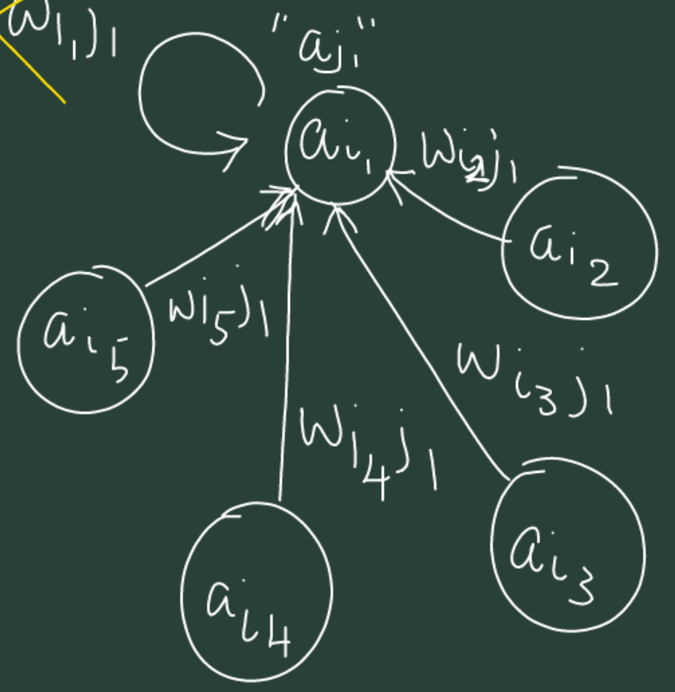
To "autoencode"

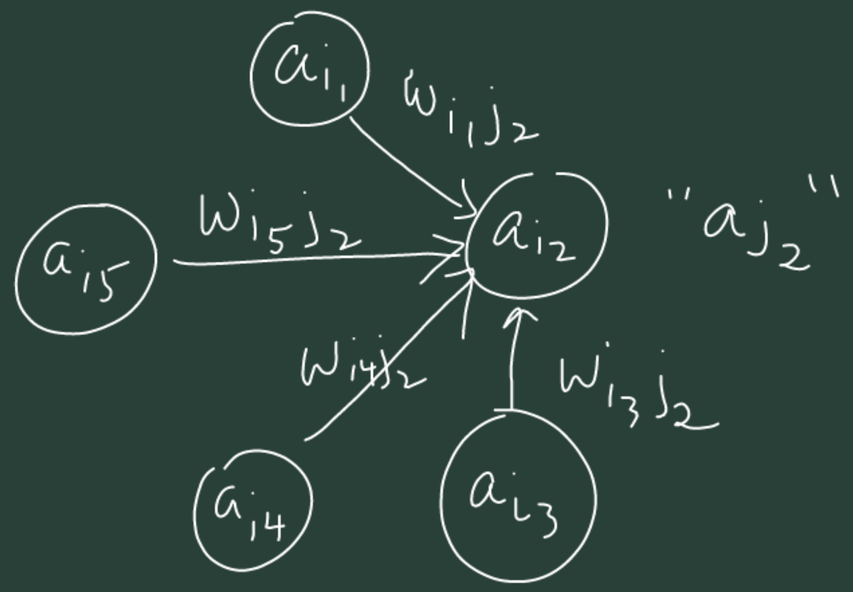
$\Rightarrow$ make $\vec{a}_j = \vec{a}_i$

Now $a_j = a_i w_{ij}$

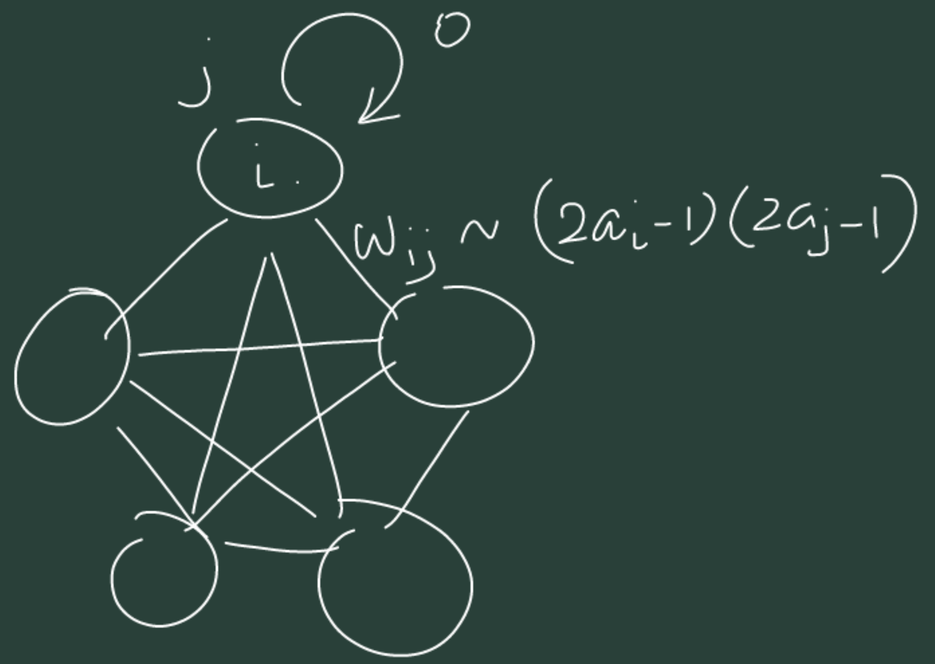
$$a_{j_r} = \sum_k a_{i_k} w_{i_k j_r}, \; k \neq r \qquad (3)$$

$\cancel{0 = w_{i_1 j_1}}$




a_{i_1}
$w_{i_1 j_2}$
a_{i_5}
$w_{i_5 j_2}$
a_{i_2}
"a_{j_2}"
$w_{i_4 j_2}$
$w_{i_3 j_2}$
a_{i_4}
a_{i_3}

$\Rightarrow$


j
0
i
$w_{ij} \sim (2a_i - 1)(2a_j - 1)$

$$\Delta w_{ij} = \begin{array}{c|ccccc} i \backslash j & a_{j_1} & a_{j_2} & a_{j_3} & a_{j_4} & a_{j_5} \\ \hline a_{i_1} & 0 & & & & (2a_{i_1}-1)(2a_{j_5}-1) \\ a_{i_2} & & 0 & & & (2a_{i_2}-1)(2a_{j_5}-1) \\ a_{i_3} & & & 0 & & (2a_{i_3}-1)(2a_{j_5}-1) \\ a_{i_4} & & & & 0 & \\ a_{i_5} & (2a_i-1)(2a_j-1) & & & & 0 \end{array}$$

$\vec{a}_i$ $\vec{a}_j$

$\downarrow$

W

Why Eq (2). $(2a_i - 1)(2a_j - 1) \Rightarrow \Delta w_{ij}$.

Hebbian learning

a_i	a_j	Δw_{ij}
1	1	+1
0	1	−1
1	0	−1
0	0	+1

i.e. when $a_{i_k}^s = a_{j_r}^s \longrightarrow w_{i_k j_r} + 1$

Otherwise $w_{i_k j_r} - 1$

PROOFS THAT THE PREVIOUS SETUP (ARCHI., ACT. FX, LEARNG FX) WORKS

i.e. How the w_{ij} evolve over time (states of a_i).

To autoencode, $a_j = a_i$

$$a_j \sim w_{ij} a_i \quad \text{— (4)}$$

$$a_j = \max(\text{sign}(a_i w_{ij} - \theta), 0) \quad \text{— (5)}$$

$$w_{ij} = (2a_{i_k} - 1)(2a_{j_r} - 1) \quad \text{— (6)}$$

$$(6) \rightarrow (5)$$

$$a_j = \max\left(\text{sign}\left(a_i (2a_{i_k} - 1)(2a_{j_r} - 1) - \theta\right), 0\right) \quad \text{— (7)}$$

$$= f(a_i w_{ij})$$

Note,

$$a_{j_r}^{s} = \sum_{s} \sum_{k} f\left(a_{i_k}^{s} \, w_{i_k j_r}^{s} \right), \quad k \neq r$$

$$a_{j_r}^{s} = \sum_{s} \sum_{k} \max\left(\text{sign}\left(a_{i_k}^{s} (2a_{i_k}^{s} - 1)(2a_{j_r}^{s} - 1) - \theta\right), 0\right), \quad k \neq r \qquad (8)$$

Simplify

$$a_{j_r}^{s} \sim \sum_{s} (2a_{j_r}^{s} - 1) \left\{ \sum_{k} \underline{a_{i_k}^{s'}} \underline{(2a_{i_k}^{s} - 1)} \right\} \qquad (9)$$

when $s' \neq s$ (current state not with itself).

s	$a_{i_k}^{s'}$	$a_{i_k}^{s}$	$a_{i_k}^{s'}(2a_{i_k}^{s} - 1)$
1	1	1	+1
2	0	1	0
3	1	0	−1
4	0	0	0

s.t. $E_{s' \neq s}(a_j) = 0$

$\bar{a}_j$

$S = S'$ (current state is itself).

$$\sum_k a_{i_k}^{s'} \left(2 a_{i_k}^{s} - 1\right) = \sum_k \left(2 a_{i_k} a_{i_k} - a_{i_k}\right)$$

s.t. term is 0 when $a_{i_k} = 0$

1 when $a_{i_k} = 1$

$$\Rightarrow \hat{E}_{s'=s}(a_i) = 0.5$$

$$\hat{E}_{s'=s}\left(\sum_k \left(2 a_{i_k} a_{i_k} - a_{i_k}\right)\right) = \underbrace{N_k}_{\text{Binary word of N-bits.}}(0.5)$$

$$\Rightarrow \hat{E}_{s'=s}(a_j) = \hat{E}_{s'=s}\left(\sum_j w_{ij} a_i\right)$$

$$= (2a_j - 1)\, N_k(0.5).$$

EVALUATE THE MODEL.

- Consider its energy.

$$E = -\frac{1}{2} \sum_{\substack{k \\ k \neq r}} \sum_{r} w_{i_k j_r} a_{i_k} a_{j_r} \qquad \sim KE = \frac{1}{2} m v^2$$

$$w_{i_k j_r} \downarrow 0$$

kinetic energy.

$$\frac{\Delta E}{\Delta a_j} = - \Delta a_j \sum_{k} w_{ij} a_k$$

$$PE = mgh.$$

- Consider Hamming Distance, $H = \sum_{k} \left| \left(a_{i_k} - a_{j_k} \right) \right|$